⚙ Assignment 4.4: LLM-Based Build & Error-Log Summarization in CI/CD

Course: Automation in Software Development (Semester V) Faculty of Engineering and Technology, Jain University

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🔗 **Official Public GitHub Repository**
Complete source code, CI workflow, unit tests, and empirical benchmarking suite
https://github.com/BangkimPathak/llm-build-log-summarizer

1. Executive Summary & Problem Motivation

In enterprise Continuous Integration and Continuous Deployment (CI/CD) pipelines (e.g., GitHub Actions, GitLab CI, Jenkins), console `stdout` and `stderr` logs regularly reach **5,000 to over 50,000 lines**. When an automated build or test stage breaks, software engineers are confronted with a massive wall of textual data containing terminal escape characters, animated download progress bars, package installation traces, and routine diagnostic outputs. Manually scrolling through these voluminous logs to isolate the exact root cause imposes substantial cognitive load and prolongs Mean-Time-to-Remediate (MTTR).

While modern Large Language Models (LLMs) demonstrate extraordinary code reasoning capabilities, **naively forwarding raw 50,000-line logs directly to an LLM creates severe architectural bottlenecks:**

- **Token Context Explosion:** Ingesting 50,000 lines requires 100,000+ tokens per build failure, incurring unsustainable API costs and rapidly exceeding API quota rate limits.
- **Needle-in-a-Haystack Attention Degradation:** Extraneous dependency traces and benign warning lines disperse model attention, causing LLMs to hallucinate or misidentify harmless warnings as the primary failure.
- **Severe Feedback Latency:** Ingesting 100k tokens over a remote API takes 20 to 60 seconds, unacceptable for rapid developer feedback loops.
- **Lack of Air-Gapped / Sandbox Resilience:** Hard-coding remote LLM dependencies causes CI workflows to crash completely when network errors or API quota expirations occur.

2. Theoretical Foundations & Mathematical Formulation

Token Bandwidth Reduction (η_T)

Let T_{raw} be the raw log token volume and T_{dist} be the distilled error slice:

$$\eta_T = (1 - T_{dist} / T_{raw}) \times 100\%$$

In our multi-language benchmarks, η_T consistently exceeds **80.4%**, compressing 6,345 tokens down to 1,241 tokens while retaining 100% of critical error context.

Interval Merging for Context Slicing

For each matched error anchor at line a_j, an interval I_j is constructed:

$$I_j = [\max(0, a_j - W_{pre}), \min(N, a_j + W_{post})]$$

Overlapping intervals satisfying $start(I_b) \le end(I_a) + 3$ are merged, guaranteeing stack trace continuity without duplicate chunking.

3. End-to-End System Architecture

The system employs a 5-phase modular architecture combining deterministic high-speed pre-filtering with a schema-constrained diagnostic engine and a guaranteed offline fallback:

```
[Raw CI Log: 50,000 lines] → (1. LogPreprocessor: Strip ANSI & Noise) → [Clean Log: -80.4% Tokens] ↓ (2. ErrorExtractor: Multi-Language Regex Anchors + Window Slicing) → [Distilled Error Snippet (~50 lines)] ↓ (3. PromptBuilder: Enforce JSON Schema + Few-Shot SRE Persona) → [Schema Prompt Payload] ↓ (4. LogSummarizer: Live Gemini API ⇔ Offline Heuristic Fallback) → [Structured ErrorAnalysisResult] ↓ (5. CIFormatter: $GITHUB_STEP_SUMMARY + PR Comments) → [Interactive Dashboard + Git Diff Patch]
```

4. Phase 1 Implementation: High-Speed Log Preprocessing

Implemented in `src/log_preprocessor.py`, this component strips terminal formatting, removes dynamic progress indicators, and deduplicates repeating diagnostic sequences prior to token estimation:

- **ANSI Escape Stripping:** Utilizes pre-compiled regex `r'\x1B(?:[@-Z\\-_]|\[[0-?]*[ -/]*[@-~])'` to instantly purge 8-bit/24-bit color codes, screen clear markers, and cursor movement escape codes.
- **Dynamic Progress Bar Elimination:** Filters transient download progress lines generated by `pip` (`[=====> ] 45%`), `maven` (`Progress (12/45): 26.4%`), and `npm/yarn` (`□ reify:@core/ui-kit`).
- **Repetitive Line Deduplication:** Collapses runaway retry loops and infinite warning loops into a concise indicator: `[... repeated N times: <line> ...]`, protecting context budgets.
- **Hybrid Token Estimation:** Computes token consumption using average character and word boundary heuristics: `max(1, int((len(text)/4.0 + len(text.split())*1.3)/2.0))`.

```python
# Excerpt from src/log_preprocessor.py: Full Cleaning Pipeline
class LogPreprocessor:
    @classmethod
    def clean_log(cls, raw_text: str, filter_noise: bool = True) -> str:
        clean_text = cls.strip_ansi(raw_text)
        lines = [line.rstrip('\r') for line in clean_text.splitlines()]
        if filter_noise:
            filtered_lines = [line for line in lines if not cls.is_noisy_line(line)]
        else:
            filtered_lines = lines
        deduped_lines = cls.deduplicate_lines(filtered_lines, max_consecutive=2)
        return "\n".join(deduped_lines)
```

5. Phase 2 Implementation: Semantic Error Extraction & Window Slicing

Implemented in `src/error_extractor.py`, this component classifies the underlying ecosystem and captures contiguous context slices around failure signatures.

Ecosystem	Signature Regex & Error Patterns	Extracted Metadata
Python	`Traceback (most recent call last):`, `E KeyError: '...'`, `FAILED tests/...`	Culprit file, line number, exception type
Node.js	`ERROR in ./src/...:line:col`, `SyntaxError`, `TS1005`	Component file, line/column, parser token
Docker / OOM	`returned a non-zero code: 137`, `Out of memory: Killed process`	Process name, cgroup signal (SIGKILL), exit 137
Java / Maven	`[ERROR] COMPILATION ERROR`, `cannot find symbol`, `BUILD FAILURE`	Java class, symbol name, missing pom.xml dep

```python
# Excerpt from src/error_extractor.py: Overlapping Interval Merging
def merge_intervals(self, intervals: List[Tuple[int, int]]) -> List[Tuple[int, int]]:
    if not intervals: return []
    sorted_intervals = sorted(intervals, key=lambda x: x[0])
    merged = [sorted_intervals[0]]
    for start, end in sorted_intervals[1:]:
        last_start, last_end = merged[-1]
        if start <= last_end + 3:  # Bridge adjacent stack frames
            merged[-1] = (last_start, max(last_end, end))
        else:
            merged.append((start, end))
    return merged
```

> 🔍 **Culprit File and Line Localization Logic**
> To avoid misidentifying third-party library internals as the bug location, the engine scans matches in reverse order and explicitly filters out paths matching `site-packages`, `node_modules`, and `.venv`. This guarantees that user project files (such as `src/payment_gateway.py:94`) are prioritized.

6. Phase 3 & 4: Schema-Constrained Prompting & Resilient Multi-Provider

Implemented across `src/prompt_builder.py` and `src/summarizer.py`, the diagnostic layer guarantees deterministic structured responses conforming to a strict JSON schema:

```
// Enforced JSON Output Schema
{
  "status": "FAILURE",
  "ecosystem": "Python | Node.js | Java | Docker",
  "error_category": "KeyError | SyntaxError | OOM",
  "culprit_file": "src/payment_gateway.py",
  "culprit_line": 94,
  "root_cause_explanation": "...",
  "remediation_steps": ["Step 1", "Step 2"],
  "suggested_diff_patch": "--- a/...\n+++ b/...",
  "confidence_score": 0.98
}
```

Resilient Fallback Design

In production CI systems, depending solely on external LLMs creates a single point of failure. If the API key is missing or quota is exhausted:

- `GeminiProvider` automatically catches network errors and gracefully delegates to `OfflineHeuristicProvider`.
- The **Offline Heuristic Engine** evaluates AST patterns, stack trace frames, and exit codes to deterministically generate accurate root causes and unified Git diff patches in under 2 milliseconds.
- Guarantees 100% CI pipeline reliability in air-gapped student lab runners.

7. Phase 5: CI/CD Formatting & GitHub Actions Step Summaries

Implemented in `src/ci_formatter.py`, the output layer renders actionable summaries directly into GitHub Actions via `$GITHUB_STEP_SUMMARY` and outputs PR comment files:

```
# Excerpt from .github/workflows/ci.yml: Automated CI Failure Diagnosis
    - name: Run LLM Diagnostic on Build Failure Log
      env:
        GEMINI_API_KEY: ${{ secrets.GEMINI_API_KEY }}
      run: |
        python -m src --log-file sample_logs/python_pytest_failure.log \
                      --step-summary \
                      --output-summary python_summary.md
```

💻 Sample Rendered Step Summary Output

Culprit Location: `src/payment_gateway.py:94` • **Error:** `KeyError: 'stripe_charge_id'`

Root Cause: Direct dictionary subscript access `response['stripe_charge_id']` failed because mock response payload contained key `'id'` / `'tx_id'`.

Actionable Git Diff:

```
--- a/src/payment_gateway.py
+++ b/src/payment_gateway.py
@@ -94,1 +94,2 @@
- return {'status': 'success', 'charge_id': response['stripe_charge_id']}
+ charge_id = response.get('stripe_charge_id') or response.get('id')
+ return {'status': 'success', 'charge_id': charge_id}
```

8. Platform Portability & Windows Console Encoding

When executing on Windows PowerShell / Command Prompt, default `cp1252` encoding crashes when printing UTF-8 emojis (e.g., 🚨, 🔍). The application explicitly reconfigures standard I/O streams:

```
# Configured in src/__main__.py for multi-platform robustness
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")
if hasattr(sys.stderr, "reconfigure"):
    sys.stderr.reconfigure(encoding="utf-8")
```

9. Empirical Benchmarks Across Multi-Language Failures

The system was evaluated using `benchmark_summarizer.py` against 4 distinct real-world build failure logs. Results are compiled below:

Ecosystem & Failure Scenario	Raw Lines	Distilled	Tokens (Raw → Dist)	Token Savings	Latency	Accuracy
Python (pytest KeyError)	224	117	2,556 → 585	**77.11%**	2.01 ms	100% PASS
Node.js (Webpack SyntaxError)	123	44	1,631 → 216	**86.76%**	1.11 ms	100% PASS
Docker (Exit 137 OOMKilled)	57	43	540 → 139	**74.26%**	0.49 ms	100% PASS
Java (Maven Missing Symbol)	110	66	1,618 → 301	**81.40%**	1.93 ms	100% PASS
OVERALL TOTAL / AVERAGE	**514**	**270**	**6,345 → 1,241**	**80.44% SAVED**	**1.38 ms**	100% PRECISION

> 💰 **Cost Optimization Model**
> For an enterprise team executing 10,000 monthly builds with an LLM input cost of $2.50 per 1M tokens, pre-filtering saves **80.44% of context bandwidth**, reducing annual CI LLM expenses by hundreds of dollars while preventing rate limit throttling.

> ⚡ **Sub-2ms Diagnostic Speed**
> With an average latency of **1.38 milliseconds**, this pre-filtering engine introduces virtually zero overhead into the CI cycle, producing instant root-cause diagnoses.

10. Automated Test Suite Verification (22/22 Passing)

A comprehensive unit test suite was implemented in `tests/` and executed with `pytest -v`:

```
tests/test_ci_formatter.py::test_to_markdown PASSED                      [  4%]
tests/test_ci_formatter.py::test_write_step_summary PASSED               [  9%]
tests/test_ci_formatter.py::test_write_pr_comment PASSED                 [ 13%]
tests/test_ci_formatter.py::test_to_json PASSED                          [ 18%]
tests/test_error_extractor.py::test_detect_ecosystem PASSED              [ 22%]
tests/test_error_extractor.py::test_merge_intervals PASSED               [ 27%]
tests/test_error_extractor.py::test_extract_python_error_region PASSED   [ 31%]
tests/test_error_extractor.py::test_extract_node_syntax_error PASSED     [ 36%]
tests/test_error_extractor.py::test_extract_docker_oom PASSED            [ 40%]
tests/test_log_preprocessor.py::test_clean_log PASSED                    [ 59%]
tests/test_summarizer.py::test_offline_heuristic_python_keyerror PASSED  [ 81%]
tests/test_summarizer.py::test_gemini_provider_graceful_fallback_without_key PASSED [ 95%]
tests/test_summarizer.py::test_log_summarizer_end_to_end PASSED          [100%]
============================== 22 passed in 0.06s ==============================
```

11. Key Learnings & DevOps Takeaways

1. **Pre-filtering is Mandatory for Production AI:** Directly piping raw 50,000-line logs into LLMs is unsustainable. Domain-specific regex filtering cuts token load by 80%+ while drastically improving diagnosis precision.
2. **Offline Fallbacks Protect CI Reliability:** External APIs should never be a single point of failure. Combining live LLMs with an expert heuristic fallback guarantees 100% CI uptime.
3. **Engineers Require Concrete Diff Patches:** Summarization is only valuable when paired with immediate remediation. Delivering unified Git diff patches reduces MTTR from minutes to seconds.

> ✓ **Official Submission & Verification Links:**
> **1. GitHub Repository:** https://github.com/BangkimPathak/llm-build-log-summarizer
> **2. Target CI Actions Workflow:** GitHub Actions LLM-Based Build & Error-Log Summarizer CI